

Super-Eddington Little Blue Dots Reionize Helium Too Early

JWST's faint AGN population may overproduce ionizing photons, conflicting with helium reionization timing.

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Background

JWST found many faint AGN at $z > 4$, called Little Blue Dots, possibly super-Eddington. Their ionizing output could affect helium reionization.

How super-Eddington accretion in Little Blue Dots impacts helium reionization timing remains unexplored.

What this paper is selling

The paper shows that if most Little Blue Dots are super-Eddington, helium reionizes too early. Observations limit the super-Eddington fraction to under 10%.

Key results to remember

- Super-Eddington LBDs would complete helium reionization before $z=3$, conflicting with data.
- At most 10% of LBDs can be super-Eddington under fiducial assumptions.
- Tighter constraints if bright quasars also contribute significantly.

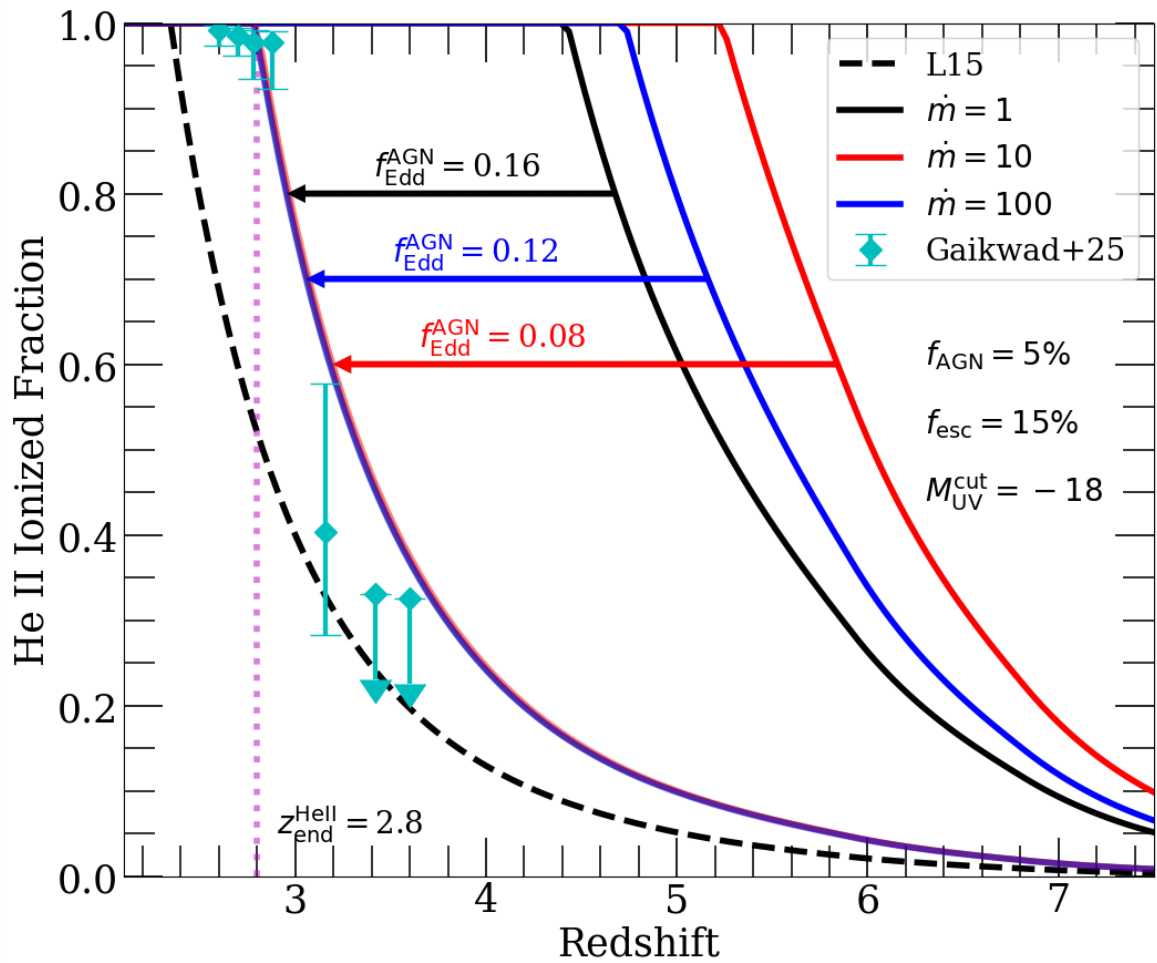


Figure 3 | He\,ii reionization histories for different SED models assuming $f_{mAGN} = 0.05$, $f_{mesc} = 0.15$, and $M_{mUV}^{mcut} = -18$.

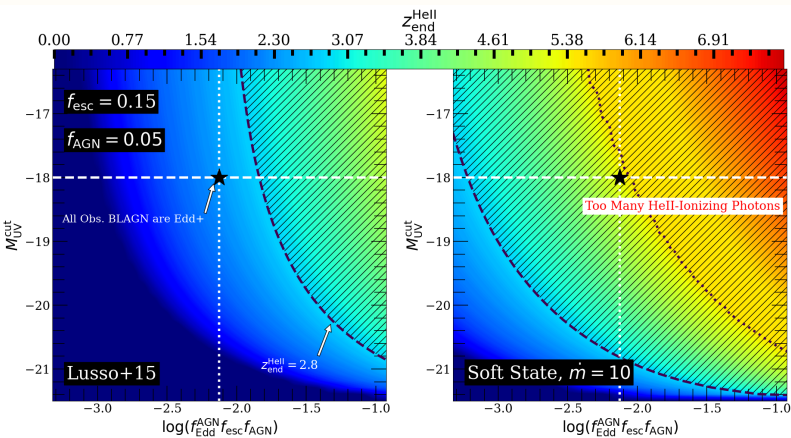


Figure 4 | AGN parameter space excluded by constraints on He\,ii reionization.